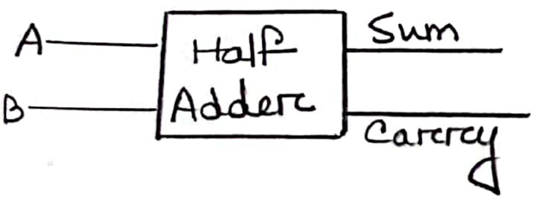
Name of the Experiment : Implimention of a full adder circuit using half adder.

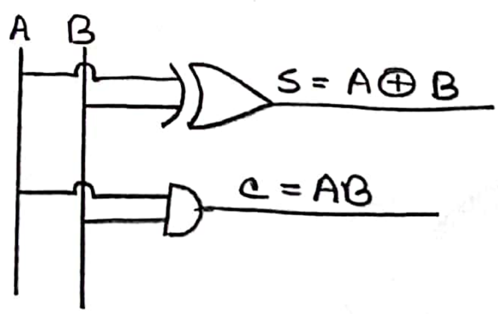
Objective :

The objective of this experiment is a full adder can be implemented by using half-adder.

Theory :

Half adder is a combinational logic circuit with two inputs and two outputs. Half adder is designed by adding two single bit binary number. This single bit adder can be easily implement with the help of X-OR gate for the output sum and an AND gate for the carry output.





Full adder :

Full adder is a combinational logic circuit with three inputs and two outputs. Full adder is developed two overcome the drow-back of half adder circuit. It adds two signal bit number A,B and a bit C carried in from the previous less significant stage.

A full adder can be constructed from two half adders and an OR gate by connecting A and B to the input of one H.A connecting the sum from that other H.A input and OR the two carry.

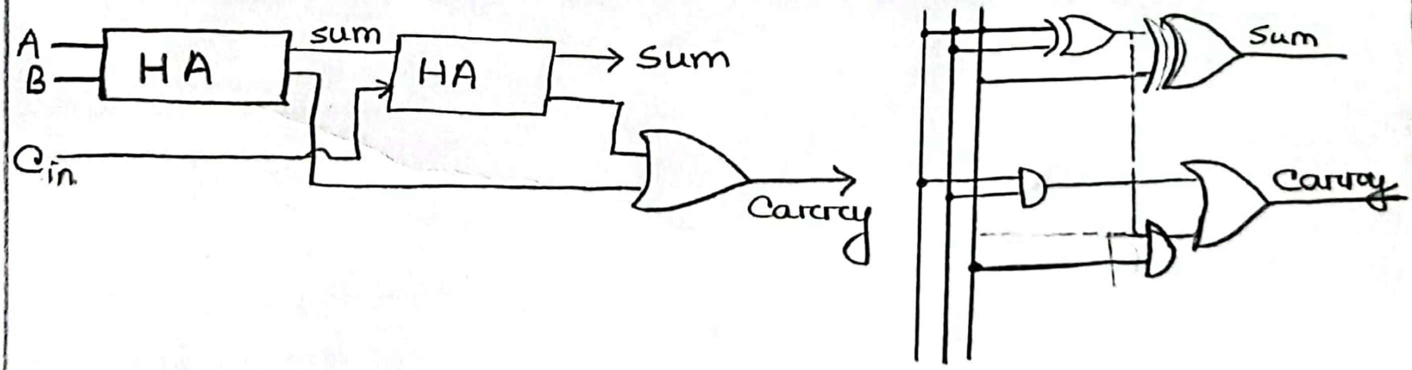


Fig: Full adder using half adder

Truth table :

Input			Output	
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	0	1
1	1	1	1	1

$Sum = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} = A \oplus B \oplus C_{in}$

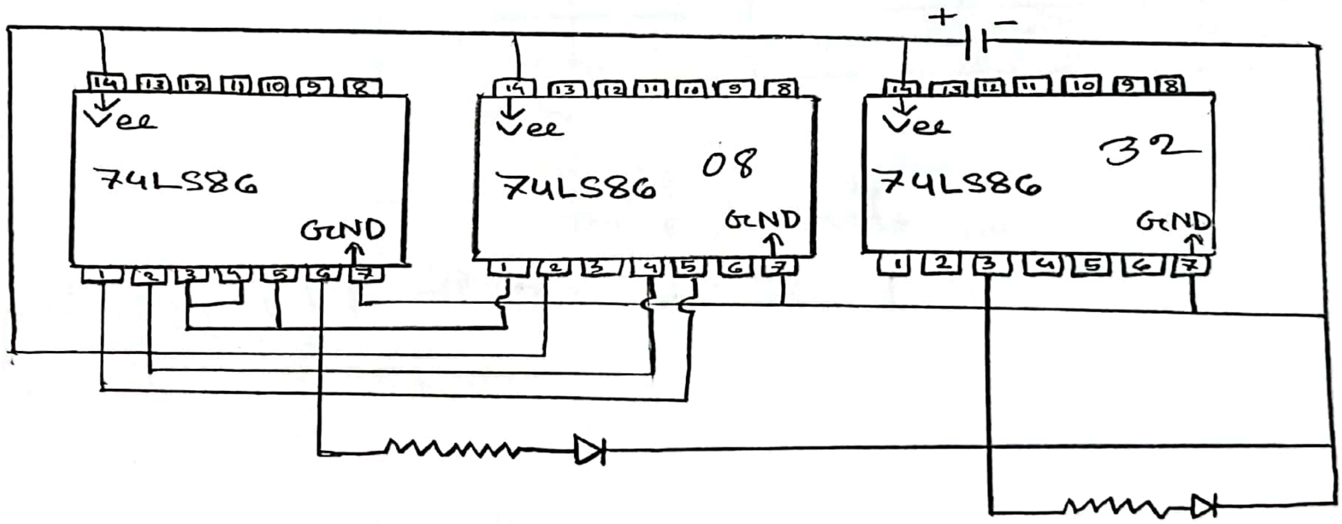
$Carry = \bar{A}BC_{in} + A\bar{B}C_{in} + ABC_{in} = AB + AC_{in} + BC_{in}$

Apparatus :

i) AND gate
ii) OR gate
iii) X-OR gate (3 input)
iv) Bread Board
v) Resistor
vi) LED
vii) Wire .

Procedure :

At first, we have to identify the pin number of the IC and then set it on the bread board. Connected all the Vcc with the positive power supply and all the ground with negative power supply. Finally connected all the input and output wires with resistor and LED.



Result:

A	B	Cin	Sum	LED	Carry	LED
0	0	0	0	OFF	0	OFF
0	0	1	1	ON	0	OFF
0	1	0	1	ON	0	OFF
0	1	1	0	OFF	1	ON
1	0	0	1	ON	0	OFF
1	0	1	0	OFF	1	ON
1	1	0	0	OFF	1	ON
1	1	1	1	ON	1	ON

Discussion:

In this experiment, we have cheaked all the gates of IC's and found the active gates. We have found out the positions where the IC's get power supply. And we connected 7 No. pin the GND and 14 No. pin in the Vcc. Then we connected all the wires in the exact point.

Thus we successfully done the experiment.

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